Jianxiong Fan

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My research examines cognitive and emotional dynamics in human–computer interaction using mixed-method approaches. I combine quantitative user experience measurement and statistical analysis with qualitative inquiry to inform data-driven product and service system design. I am particularly interested in extending this work to AI-supported learning and human–AI collaboration.

Education

Guangdong University of Technology (GDUT) *Sept 2023- Jun 2026*

M.A. in Design Studies | Digital Creative Design Research

- **Academic score:** 91.1/100 (top 10%); **Thesis defence score:** 92.2/100.
- **Supervisor:** Prof. Chaolan Tang; **Co-supervisor:** Assoc. Prof. Xian Yang.
- **Honours:** GDUT Elite Innovation Talent; Outstanding Master's Graduate.
- **Core Courses:** Design Psychology Research; User Research; Service Design Research

Guangdong University of Technology (GDUT) *Sept 2019 - Jun 2023*

B.Eng. in Digital Media Technology

- **Academic score:** 88.86/100 (Rank: 3rd out of 58)
- **Honours:** Outstanding Bachelor's Graduate (top 3%)
- Granted eligibility for postgraduate admission by recommendation, exempt from the national entrance examination.
- **Core Courses:** Virtual Reality Technology; 3D Animation Technology Practice; C Language Programming

Publications and Manuscripts

A. Published journal articles:

[A1] **Jianxiong Fan**, Rouqin Wu, Chaolan Tang, & Xian Yang. (2026). Empowering Vocational Training for Middle-aged, Elderly, and Low-educated Individuals: A Design Approach for Entertainment-Based Learning. *International Journal of Human–Computer Interaction*. DOI:10.1080/10447318.2025.2536624 **(JIF(2025) = 6.1, SCI/SSCI Q1)**

[A2] **Jianxiong Fan** et al. (2026). Key Elements of Short-Video Entertainment-Based Learning for Vocational Training in Middle-Aged and Older Adults with Low Educational Attainment. *Thinking Skills and Creativity*. DOI: 10.1016/j.tsc.2026.102295 **(JIF(2025) = 5.3, SSCI Q1)**

[A3] Rouqin Wu, **Jianxiong Fan**, Chaolan Tang, Xian Yang (2025). Interaction-based creative thinking research and its application in artistic creation. *Thinking Skills and Creativity*, 56, 101750. DOI:10.1016/j.tsc.2024.101750 **(JIF(2025) = 5.3, SSCI Q1)**

[A4] Rouqin Wu, **Jianxiong Fan** et al. (2026). Creative Thinking in Artistic Creation: Cognitive and Emotional Mechanisms of Interaction-based Thinking. *Thinking Skills and Creativity*. DOI: 10.1016/j.tsc.2026.102297 **(JIF(2025) = 5.3, SSCI Q1)**

B. Manuscripts:

[B1] **Jianxiong Fan** et al. From Psychophysiological Process Fit to Technology Acceptance: A Dynamic Adaptation Mechanism in Short-Video-Based Safety Training. *Under review at Information Processing & Management.*

[B2] **Jianxiong Fan** et al. Reconceptualising Cognitive Load in Short-Video Vocational Training: A Three-Dimensional Framework Supported by Psychophysiological and Behavioural Evidence. *Manuscript in preparation.*

[B3] Rouqin Wu, **Jianxiong Fan** et al. A Study on the observable behavioral differences of different types of creative thinking in artistic creation. *Submitted to The Journal of Creative Behavior.*

Research Experiences

Short-Video Entertainment-Based Learning Design

Feb 2022 – Present

Funding: Design and Development of a Defensive Driving Training System (Project No. 23HS0022)

Overview: Investigated the design and mechanisms of short-video entertainment-based learning for hazardous chemical transport drivers, examining age- and education-related needs alongside learners' cognitive, emotional and behavioural responses.

Key Responsibilities & Contributions:

- Conducted qualitative research, factor analysis and K-means clustering to identify learner characteristics and training needs, and proposed an entertainment-based approach to short-video learning.
- Led the design of a short-video training app, an integrated backend management system and an immersive VR-based assessment platform.
- Conducted a BIOPAC-based psychophysiological experiment with 48 target-group drivers and 48 younger university participants, finding stronger electrodermal responses and higher preference ratings for short videos than for text-and-graphics materials.
- Analysed post-training survey data (N = 333) from a six-month programme using an integrated TPB–TAM model, identifying positive associations of perceived usefulness and attitudes towards training with participation intention (model R^2 = .516).
- Under academic supervision, proposed a three-dimensional cognitive-load framework for short-video learning, incorporating intrinsic, dual-valence extraneous and emotion-related cognitive load.

Related Outcomes:

- Two first-author journal articles [A1, A2], one manuscript under review [B1] and one manuscript in preparation [B2]. This work formed the basis of my master's thesis.
- The system was integrated with a partner enterprise in Fujian. Project outputs supported one national and four provincial competition awards.

Mechanisms of Interaction-Based Creative Thinking

Sept 2023 – Present

Funding: Guangdong Basic and Applied Basic Research Foundation under Grant 2026A1515012303

Overview: Investigates cognitive, emotional and behavioural processes in artistic collage, comparing interaction-based and divergence-based creative thinking and examining the roles of text–image semantic relatedness and emotional text.

Key Responsibilities & Contributions:

- Contributed to literature review, experimental design, think-aloud protocol development, cognitive and behavioural coding, and lag sequential analysis.
- Contributed to studies showing higher cognitive-depth scores for interaction-based than divergence-based thinking when text and images were semantically related.
- Co-authored two published journal articles [A3, A4] and contributed to a submitted manuscript [B3].
- Led an EEG and eye-tracking follow-up study on emotional text; designed experimental materials, procedures and multimodal data collection protocols, and analysed EEG and eye-tracking data to examine cognitive and emotional processing.

Research and Technical Skills

Methods: Mixed-methods research; field research and interviews; questionnaire-based research; factor analysis; cluster analysis; structural equation modelling (SEM); think-aloud protocols; behavioural coding; lag sequential analysis.

Analysis tools: SPSS, AMOS, SmartPLS and MATLAB.

Experimental methods: EEG/ERP; Eye-tracking; Multichannel physiological recording.

Design and prototyping: Unity-based AR/VR interaction; Figma, Adobe Photoshop; Premiere Pro; Maya and 3ds Max.

Additional Research & Technical Experience

Lab Assistant, Human-Centered Technology Lab *Mar 2024 — Mar 2026*

Key Responsibilities: Gained practical experience in operating EEG, eye-tracking and multichannel physiological recording systems; maintained and managed laboratory equipment as part of routine laboratory operations.

Research Exchange with Macao University of Science and Technology (MUST) *Jun — Aug 2024*

Key Responsibilities: Led laboratory tours and explained the operating principles and research applications of Tobii Pro Nano and BIOPAC MP150. Delivered an invited presentation at MUST on the team's applications of structural equation modelling.

International Workshop on AR/VR Interaction Design *Nov 2021 — Apr 2023*

Advisors: Prof. Benlin Du, The Hong Kong Polytechnic University; Assoc. Prof. Ji Yi, GDUT.

Key Responsibilities: Led a university-level student innovation and entrepreneurship training project to develop an interactive 3D exhibition system for Foshan New Year Posters. Conducted field interviews, affinity mapping, user surveys and service journey redesign; co-developed high-fidelity UI prototypes and Unity-based AR/VR modules for craft simulation and quiz interaction.

Outcomes: Recognised as an Outstanding Student. Related project outputs included two national and three provincial competition awards and two software copyright registrations.

Park Accessibility in Age-Friendly Cities *Jan — Dec 2022*

Funding: 2021 Guangdong Provincial Science and Technology Innovation Strategy Special Fund

Key Responsibilities: Contributed to a team study of accessibility in Guangzhou's parks and green spaces, focusing on older adults and people with disabilities. The study combined field observation, interviews, surveys and participatory action research to develop recommendations for improving park accessibility.

Team Outcomes: Third Prize at the national finals of the "Challenge Cup" competition; nine reports by national and provincial media outlets; accessibility design guidelines and accompanying illustrations for parks and green spaces.

Selected Scholarships & Honors

2022, 2025	National Scholarship, Ministry of Education, China. Two awards; CNY 8,000 (2022) and CNY 20,000 (2025).
2020, 2021	National Encouragement Scholarship, Department of Education of Guangdong Province. CNY 5,000 per award.
2023-2025	First-Class Academic Scholarship for Master's Students, GDUT. Three awards; CNY 10,000 per award.
2025	37 Interactive Entertainment Corporate Scholarship. CNY 4,000 per award.
2022	First-Class Outstanding Student Scholarship, GDUT. CNY 2,000 per award.
2020, 2021	Second-Class Outstanding Student Scholarship, GDUT. Two awards; CNY 1,000 per award.
2020, 2022	Academic Merit Scholarship, GDUT. Two awards; CNY 500 per award.

Selected Software Copyrights

Registered with the National Copyright Administration of China.

[S1] Jianxiong Fan et al. Foshan New Year Posters Museum. Registration No. 2025SR0798459.

[S2] Jianxiong Fan et al. New Year Painting World Software. Registration No. 2025SR0738059.

[S3] Jianxiong Fan et al. Mathematical Puzzle VR Game. Registration No. 2024SR1834354.

References

Prof. Chaolan Tang | Guangdong University of Technology | tangchl@gdut.edu.cn

Assoc. Prof. Xian Yang | Guangdong University of Technology | yangxian@gdut.edu.cn